Syllabus of AAE 42100-02

Flight Dynamics and Control

Spring 2026

Course Information

- Course number and title: AAE 42100 Flight Dynamics and Control
- CRN: 10597
- Meeting time and location: M/W 4:30 pm-5:45 pm at Forney Hall of Chemical Engr B124
- Course credit hours: 3
- Course Brightspace page: https://purdue.brightspace.com/d2l/home/1481880
- Prerequisites: AAE 34000 and AAE 36400

Instructor Contact Information

- Name of the instructor: Dr. Ran Dai
- Office Location: ARMS 3227
- Phone number: 765-496-5243
- Purdue Email Address: randai@purdue.edu

Course Description

Flight vehicle rigid-body equations of motion; linearization via small perturbation techniques. Trim analysis, static and dynamic stability, aerodynamic stability derivatives and control effectiveness. Vehicle transfer functions, stability augmentation, aircraft handling qualities. Introduction to flexible vehicle effects.

Learning Resources, Technology & Texts

- Required textbook: Flight Dynamics and Control of Aero and Space Vehicles, by Rama Yedavalli, 2020, ISBN: 978-1-118-93445-6
- Reference book: Flight stability and automatic control, 2nd Edition, by Robert C. Nelson, 1998.
- Software: Matlab or Python, Simulink (free for all Purdue students)
- Brightspace page: All course-related materials will be available at Brightspace.
- Gradescope: All assignments MUST be submitted through gradescope (Entry Code: **V4W2R7**)
- Piazza Signup-List: https://piazza.com/purdue/spring2026/aae42100section2
- Use of artificial intelligence (AI) or Large Language Models (LLM) in this course: Students shall not use AI tools during in-class examinations, group project, or assignments unless explicitly permitted and instructed.
- Academic Support: The Helen Bass Williams Academic Success Center, provides a variety of proactive, practical and approachable academic support services for you to strengthen your approaches and strategies for learning, including study skills consultations, peer coaching,

workshops, and online handouts. Visit the ASC website for more information and to access resources.

Learning Outcomes

On completing this course, the student shall be able to:

1. Develop state space and transfer function models of physical systems
2. Simulate nonlinear systems
3. Linearize nonlinear systems
4. Analyze a system for stability
5. Design a feedback controller to meet stability and performance requirements
6. Analyze a flight vehicle for trim conditions, stability and handling qualities

Assignments

Your learning will be assessed through a combination of homework, two middle term exams, a group project, and a final exam spread throughout the academic period. Details on these assignments and exams, *including* a schedule of due dates, rubrics to guide evaluation, and guidelines on discussion participation and evaluation will be posted on the course website throughout the semester. Assignments submission links will be provided at Gradescope.

Assignments	Percentage
Homework	25%
Two middle term Exams	30% (15% for each one)
Group Project	15%
Final Exam	30%
	Total: 100%

Grading Scale

In this class, grades reflect the sum of your achievements throughout the semester. You will accumulate points as described in the assignments portion above, with each assignment counted according to the specified percentage. At the end of the semester, final grades will be calculated by adding the total points earned and translating those numbers (out of 100) into the following letters (**there will be no partial points or rounding**).

Letter Grade	Percentage	Performance
A	93-100%	Excellent Work
A-	89-92.99%	Nearly Excellent Work
B+	86-88.99%	Very Good Work
B	81-85.99%	Good Work
B-	79-80.99%	Mostly Good Work
C+	76-78.99%	Above Average Work
C	71-75.99%	Average Work
C-	69-70.99%	Mostly Average Work
D+	66-68.99%	Below Average Work
D	61-65.99%	Poor Work
F	0-60.99%	Failing Work

NOTE: The percentage weights on the exams, scales, projects, and homework may be adjusted to produce the best final grade.

Course Schedule (Tentative*)

Week	Topic & Readings	Assignments
Week 1	Introduction and Rigid Body Dynamics (Coordinates, Position, Velocity)	
Week 2	Rigid Body Dynamics (Euler Forces and Moments)	Homework 1
Week 3	Orientation, Euler Angle and Flight Dynamics Equations of Motion	Homework 2
Week 4	Static Analysis, Lift & Pitch Moment of Tail	Homework 3
Week 5	Linearization & Perturbation, Linear System Solutions	Homework 4
Week 6	Review 1 & Exam 1	Middle Exam 1
Week 7	Introduction of Simulink and Simulink Solutions	Homework 5
Week 8	Aircraft Dynamics Model Decomposition and Approximation	Homework 6
Week 9	Classical Control Design	
Week 10	Spring Break	
Week 11	Classical Control Design	Homework 7
Week 12	Modern Control Design	
Week 13	Modern Control Design	Homework 8
Week 14	Review 2 & Exam 2	Middle Exam 2
Week 15	Case Studies	
Week 16	Group Project Presentation	Project Report

* Schedule and assignments are subject to change.